

Class Activity – 14/08/2026

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1. A university wants to determine whether students who spend more hours studying obtain higher marks.

```
StudyHours <- c(2, 3, 4, 5, 6, 7, 8, 9)
```

```
Marks <- c(45, 52, 58, 63, 69, 74, 81, 88)
```

```
student_data <- data.frame(StudyHours, Marks)
```

```
print(student_data)
```

```
plot(student_data$StudyHours,
```

```
student_data$Marks,    main =
```

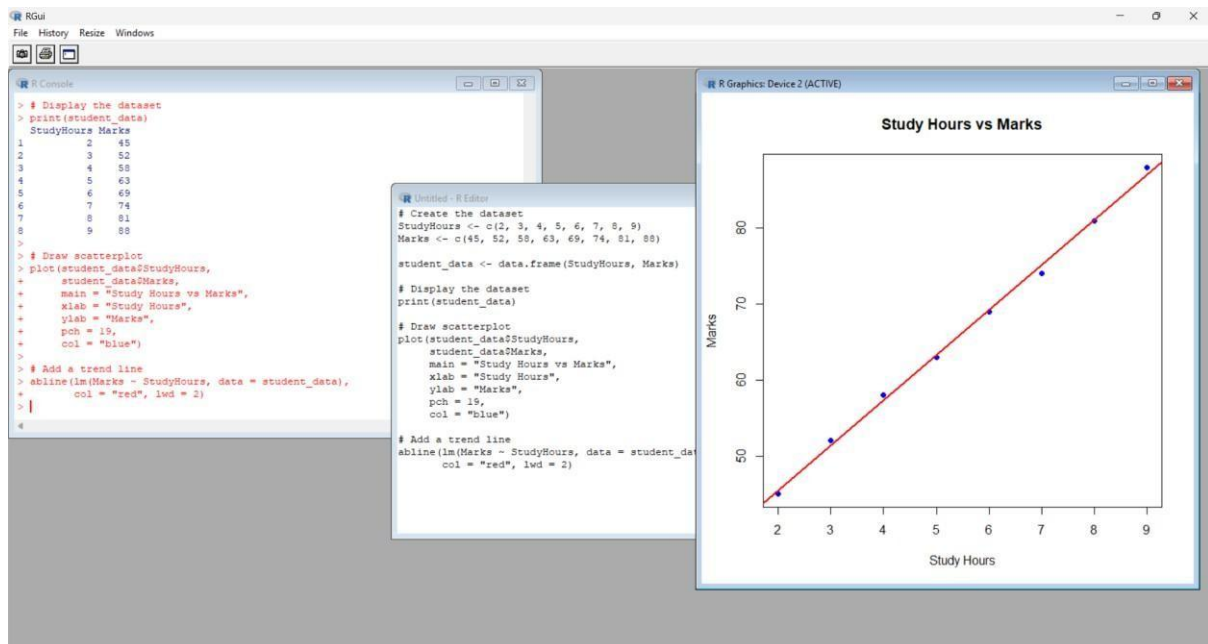
```
"Study Hours vs Marks",    xlab =
```

```
"Study Hours",    ylab = "Marks",
```

```
pch = 19,    col = "blue")
```

```
# Add a trend line abline(lm(Marks ~ StudyHours,
```

```
data = student_data),    col = "red", lwd = 2)
```



2. A hospital records patients' age and systolic blood pressure.

Python Code:

```
import matplotlib.pyplot as plt
```

```
age = [25, 30, 35, 40, 45, 50, 55, 60] bp =
[110, 115, 118, 122, 128, 135, 140, 148]
```

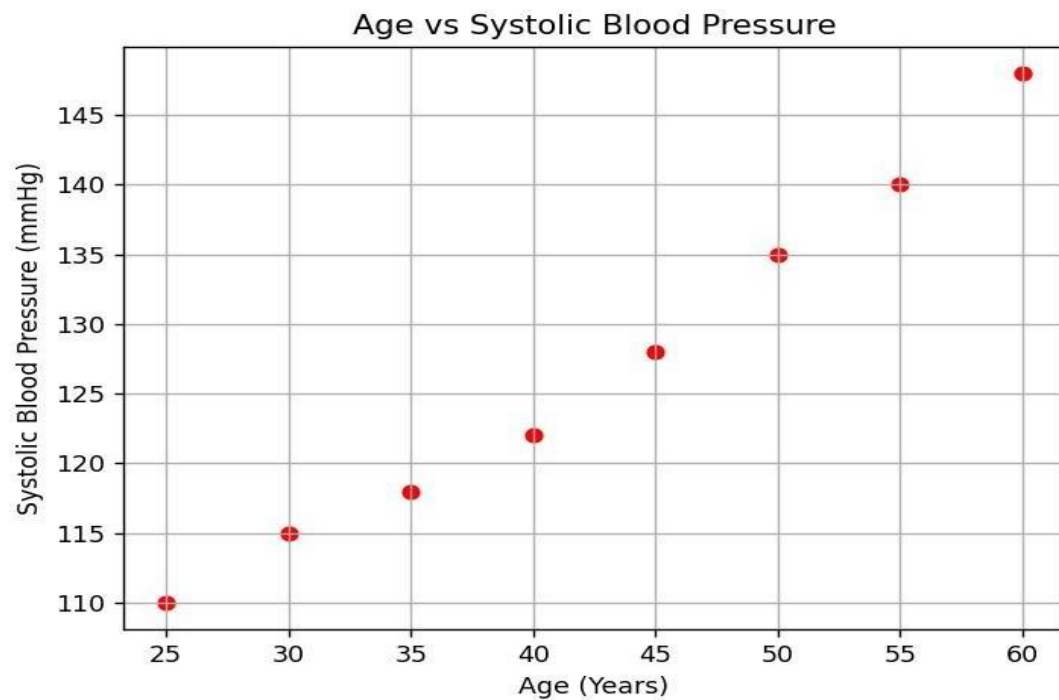
```
plt.scatter(age, bp, color='red')
```

```
plt.title("Age vs Systolic Blood Pressure")
```

```
plt.xlabel("Age (Years)") plt.ylabel("Systolic
Blood Pressure (mmHg)")
```

```
plt.grid(True)
```

```
plt.show()
```

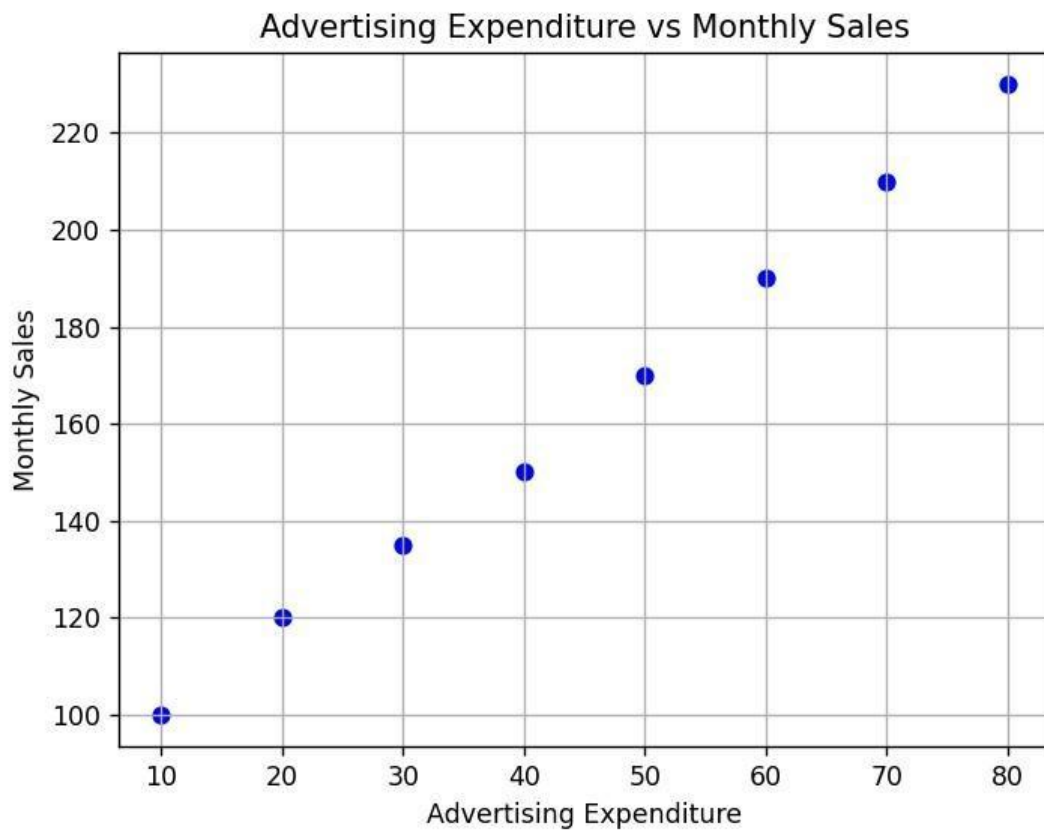


3. A company wants to investigate whether advertising expenditure influences monthly sales. Develop a scatterplot and determine whether sales increase with advertising expenditure.

```
import matplotlib.pyplot as plt
```

```
advertising = [10,20,30,40,50,60,70,80] sales  
= [100,120,135,150,170,190,210,230]
```

```
plt.scatter(advertising, sales, color='blue')  
plt.title("Advertising Expenditure vs Monthly Sales")  
plt.xlabel("Advertising Expenditure")  
plt.ylabel("Monthly Sales") plt.grid(True) plt.show()
```



4. A financial institution has collected information on:

```
import pandas as pd
```

```
import seaborn as sns
```

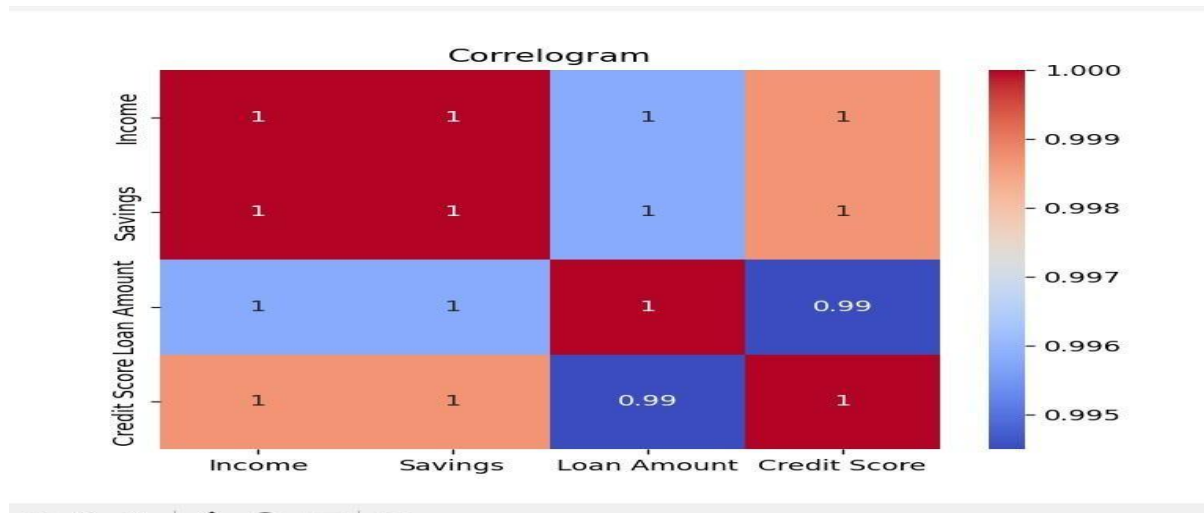
```
import matplotlib.pyplot as plt
```

```
data = {  
    'Income': [30000, 35000, 40000, 45000, 50000, 55000],  
    'Savings': [5000, 7000, 9000, 11000, 13000, 15000],  
    'Loan Amount': [10000, 12000, 15000, 18000, 21000, 25000],  
    'Credit Score': [620, 650, 680, 700, 730, 760]  
}
```

```
df = pd.DataFrame(data)
```

```
corr = df.corr()  
print(corr)
```

```
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlogram") plt.show()
```



5. Using the Iris dataset,

Compute the correlation matrix.

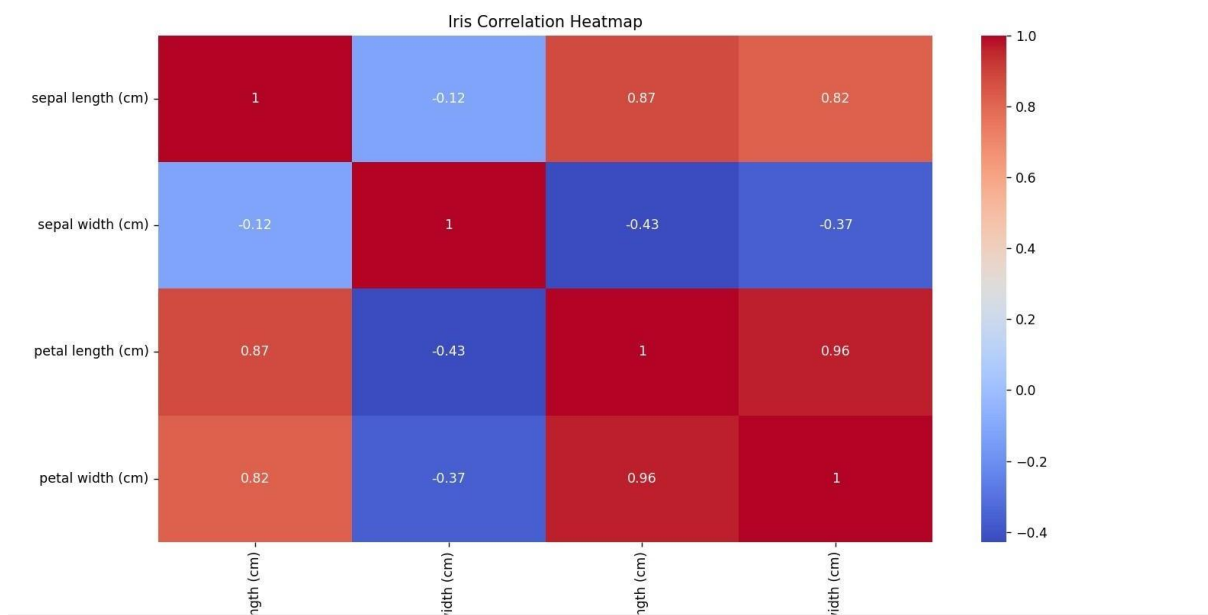
Display it as a heatmap using Seaborn.

Identify variables having negative correlation.

```
from sklearn.datasets import load_iris
import pandas as pd
import seaborn
as sns import matplotlib.pyplot as plt
```

```
iris = load_iris()
```

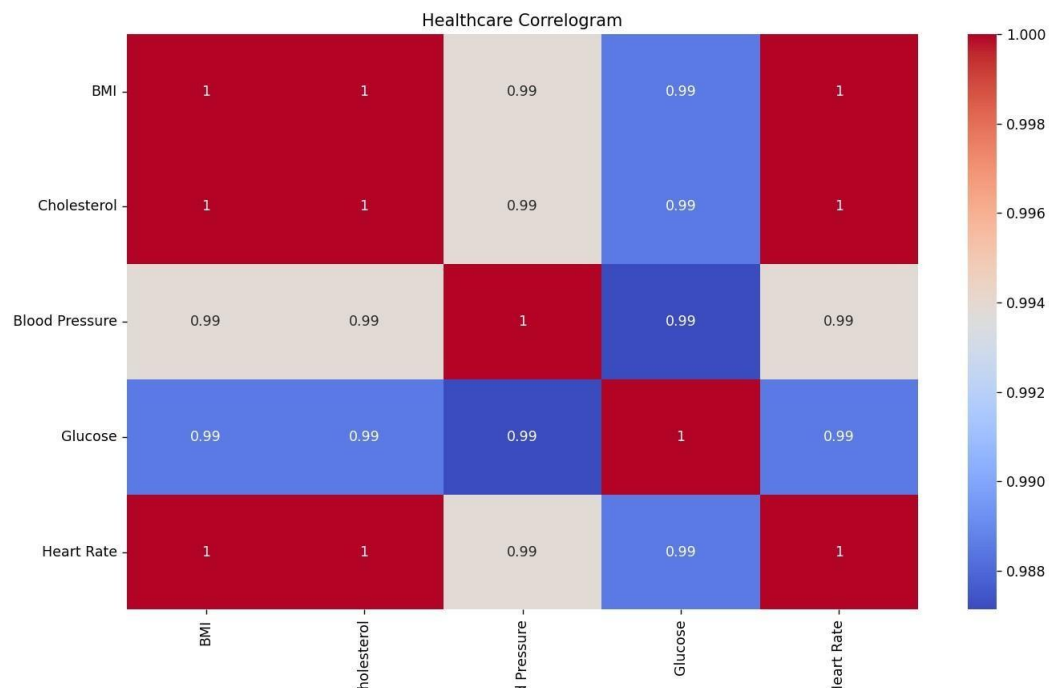
```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
corr = df.corr() print(corr)
sns.heatmap(corr, annot=True,
cmap='coolwarm') plt.title("Iris Correlation Heatmap")
plt.show()
```



6. A healthcare dataset

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data = {
    'BMI': [22, 24, 26, 28, 30, 32],
    'Cholesterol': [180, 190, 200, 210, 220, 230],
    'Blood Pressure': [110, 115, 120, 130, 135, 140],
    'Glucose': [90, 95, 100, 110, 120, 130],
    'Heart Rate': [70, 72, 74, 76, 78, 80]
}
df = pd.DataFrame(data)
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Healthcare Correlogram")
plt.show()
```



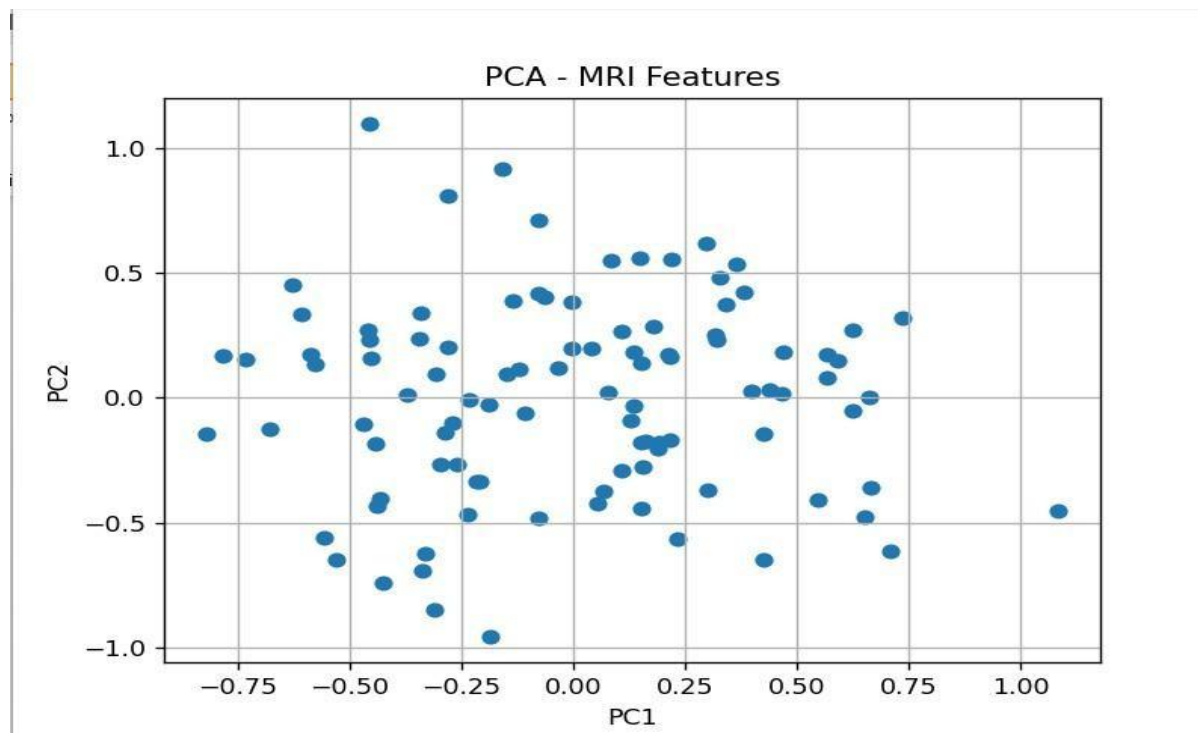
7. A researcher is analyzing 25 features extracted from MRI images. Perform PCA to reduce the dimensionality to two principal components. Plot the transformed data.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
```

```
X = np.random.rand(100,25)
```

```
pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X)
plt.scatter(X_pca[:,0],X_pca[:,1])
plt.title("PCA - MRI Features")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.grid(True)
plt.show()
```



8. Using the Iris dataset from

```
sklearn.datasets import load_iris from
```

```
sklearn.decomposition import PCA
```

```
import matplotlib.pyplot as plt
```

```
iris = load_iris()
```

```
X = iris.data
```

```
pca = PCA()
```

```
X_pca = pca.fit_transform(X)
```

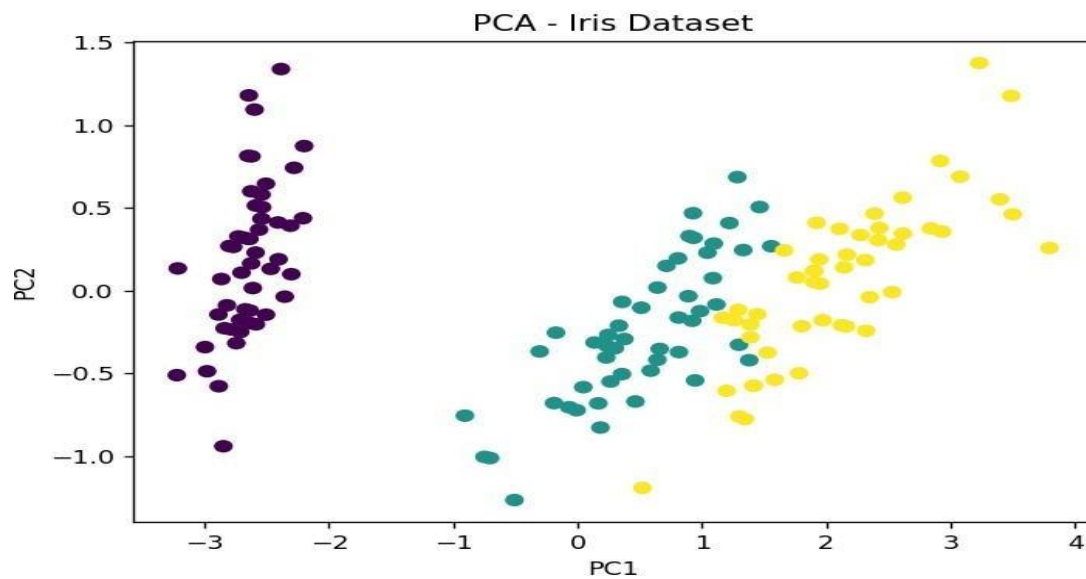
```
print("Explained Variance Ratio")
```

```
print(pca.explained_variance_ratio_)
```

```
plt.scatter(X_pca[:,0],X_pca[:,1],c=iris.target)
```

```
plt.xlabel("PC1") plt.ylabel("PC2")
```

```
plt.title("PCA - Iris Dataset") plt.show()
```

9. An automobile company has recorded 15 specifications for various cars. Reduce the dimensions using PCA and visualize the first two principal components.

```
import numpy as np from
```

```
sklearn.decomposition import PCA
```

```
import matplotlib.pyplot as plt
```

```
X = np.random.rand(100,15)
```

```
pca = PCA(n_components=2)
```

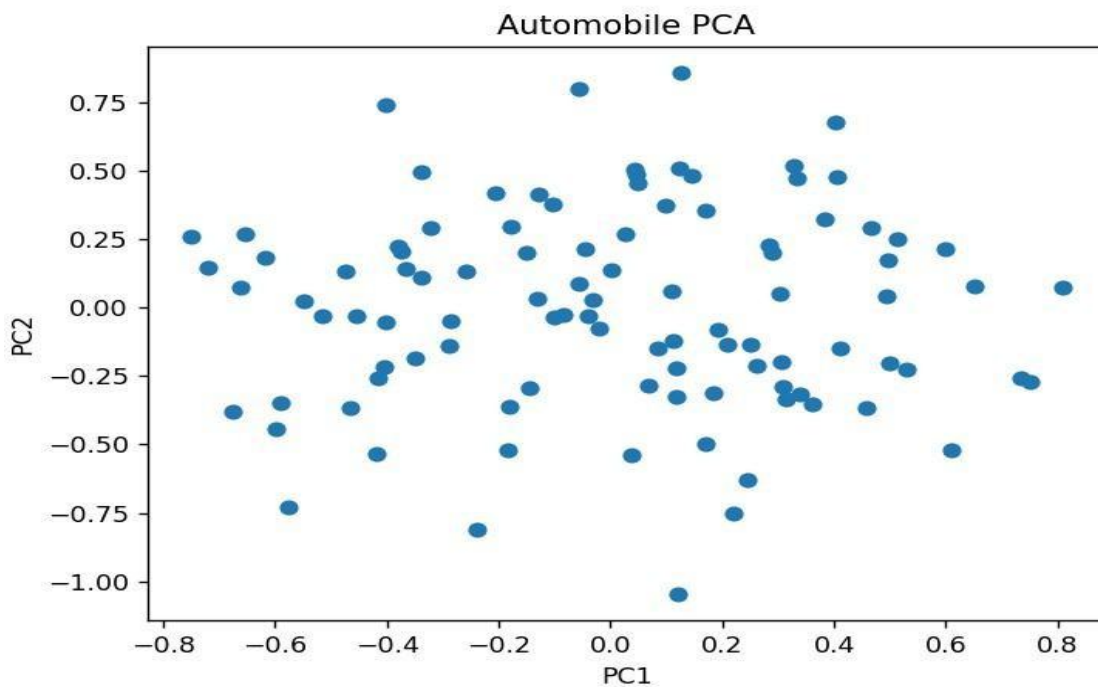
```
result = pca.fit_transform(X)
```

```
plt.scatter(result[:,0],result[:,1])
```

```
plt.title("Automobile PCA")
```

```
plt.xlabel("PC1")
```

```
plt.ylabel("PC2") plt.show()
```



10. A facial recognition dataset contains 128-dimensional facial embeddings.

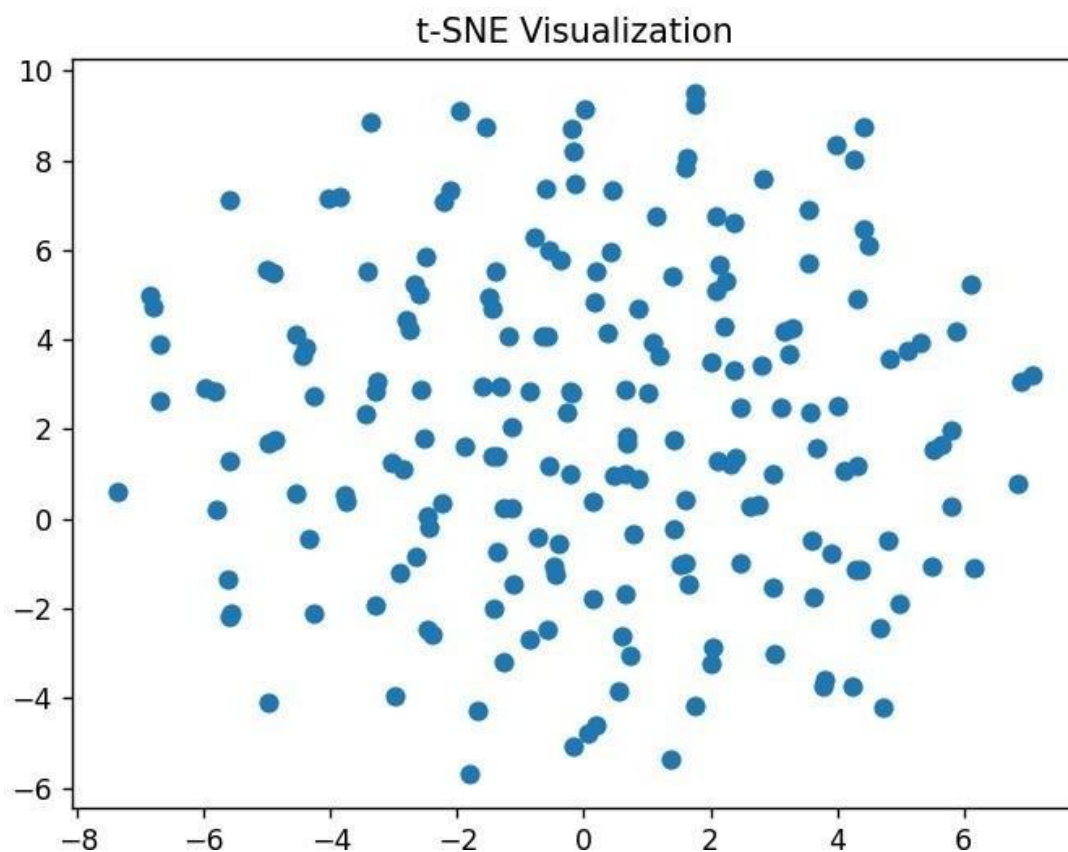
```
import numpy as np from
sklearn.manifold import TSNE
import matplotlib.pyplot as plt

X = np.random.rand(200,128)

tsne = TSNE(n_components=2, random_state=42)

X_tsne = tsne.fit_transform(X)

plt.scatter(X_tsne[:,0],X_tsne[:,1])
plt.title("t-SNE Visualization") plt.show()
```



11. Using the Iris dataset,

Apply t-SNE and plot the resulting clusters using different colors for each species.

```
from sklearn.datasets import load_iris from  
sklearn.manifold import TSNE import  
matplotlib.pyplot as plt
```

```
iris = load_iris()
```

```
X = iris.data y
```

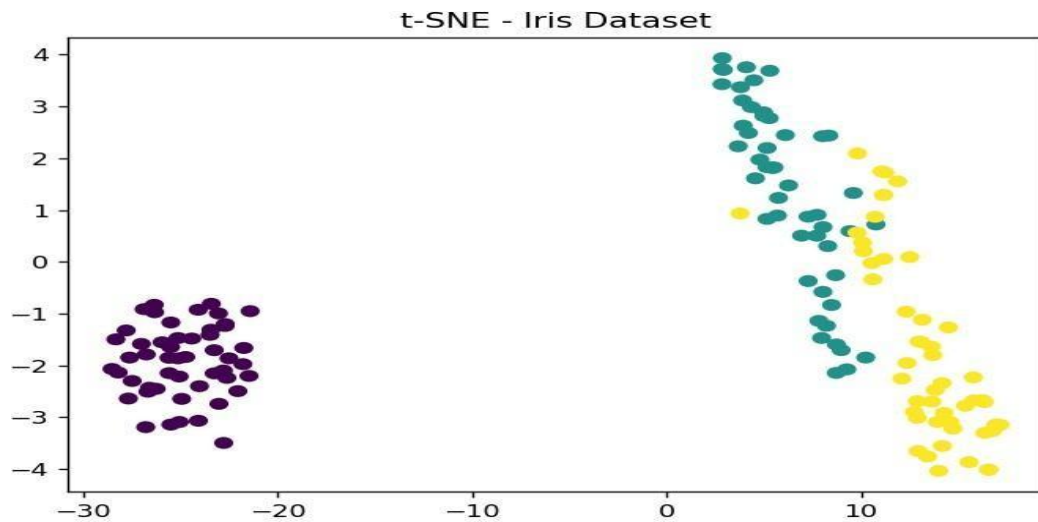
```
= iris.target
```

```
tsne = TSNE(n_components=2, random_state=42)
```

```
X_tsne = tsne.fit_transform(X)
```

```
plt.scatter(X_tsne[:,0],X_tsne[:,1],c=y)
```

```
plt.title("t-SNE - Iris Dataset") plt.show()
```



12. A customer segmentation dataset contains 40 behavioral variables.

```
import numpy as np from
```

```
sklearn.manifold import TSNE
```

```
import matplotlib.pyplot as plt
```

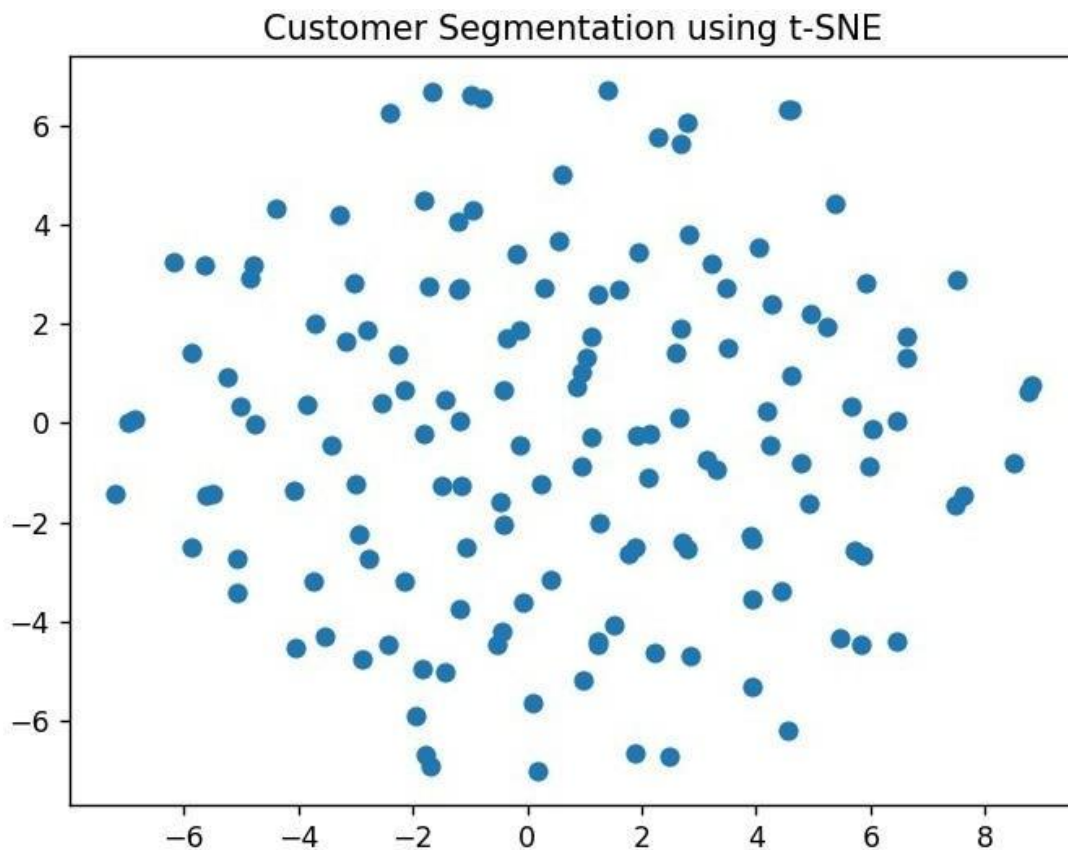
```
X = np.random.rand(150,40)
```

```
tsne = TSNE(n_components=2, random_state=42)
```

```
result = tsne.fit_transform(X)
```

```
plt.scatter(result[:,0],result[:,1]) plt.title("Customer
```

```
Segmentation using t-SNE") plt.show()
```



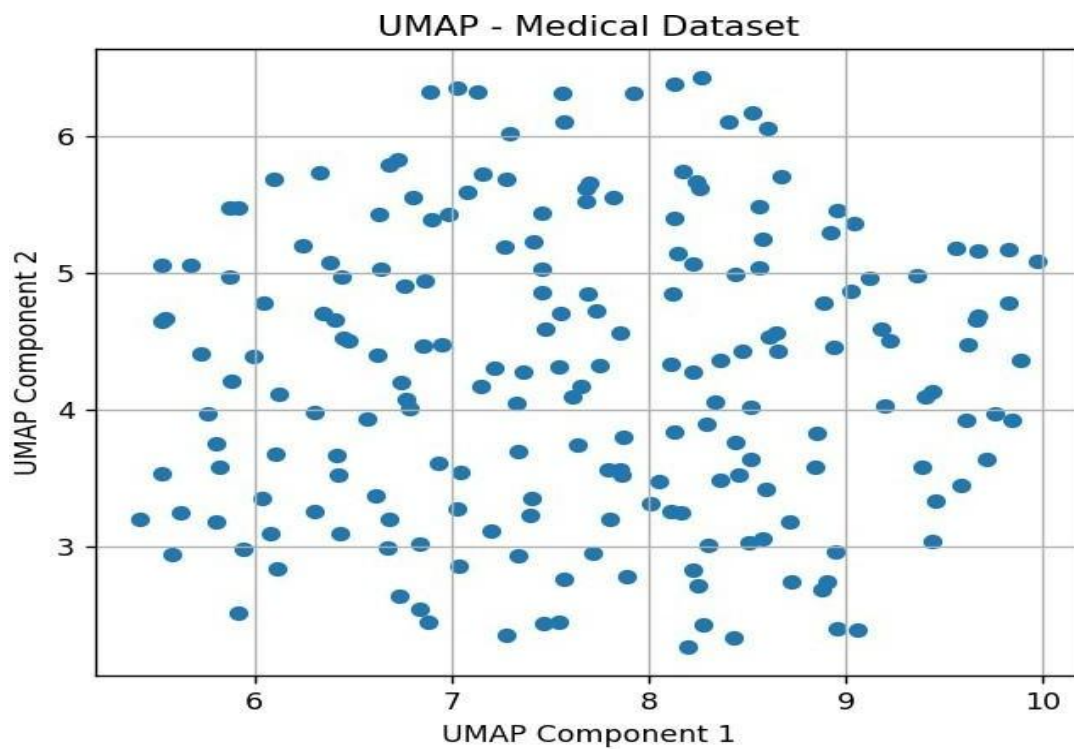
13. A medical image dataset contains 100 extracted features.

```
import numpy as np
import matplotlib.pyplot as plt
from umap import UMAP

X = np.random.rand(200, 100)
reducer = UMAP(n_components=2, random_state=42)

embedding = reducer.fit_transform(X)

plt.figure(figsize=(6,5))
plt.scatter(embedding[:, 0], embedding[:, 1], s=30)
plt.title("UMAP - Medical Dataset")
plt.xlabel("UMAP Component 1")
plt.ylabel("UMAP Component 2")
plt.grid(True)
plt.show()
```



14. Apply UMAP to the Iris dataset.

Plot the reduced dimensions using different colors for each species.

```
from sklearn.datasets import load_iris
```

```
import matplotlib.pyplot as plt import
```

```
umap
```

```
iris = load_iris()
```

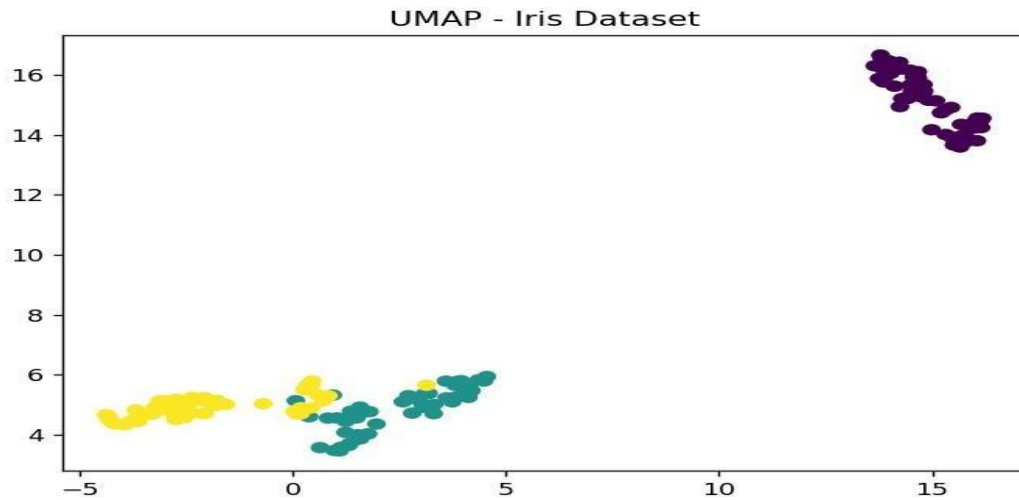
```
X = iris.data y
```

```
= iris.target
```

```
reducer = umap.UMAP(random_state=42)
```

```
embedding = reducer.fit_transform(X)
```

```
plt.scatter(embedding[:,0],embedding[:,1],c=y)
plt.title("UMAP - Iris Dataset") plt.show()
```



15. An e-commerce company has collected customer purchasing behavior consisting of 60 attributes.

```
import numpy as np
import matplotlib.pyplot as plt
import umap
```

```
X = np.random.rand(200,60)
```

```
reducer = umap.UMAP(random_state=42)
```

```
embedding = reducer.fit_transform(X)
```

```
plt.scatter(embedding[:,0],embedding[:,1])
plt.title("Customer Purchasing Behavior - UMAP")
plt.show()
```

